CASSARIA COMMODITY

CasCom

Quantitative Portfolio Research

Constructed April 14, 2025 | Capped Risk Parity (CRP) Weighting

+52.7%	+24.0%	2.54	0.50
Live Return	Alpha vs SPY (Live)	OOS Sharpe	Beta vs SPY

Analysis Period: April 2020 — April 2025 | Live: April 2025 — March 2026

Dublin, 2026

Table of Contents

1. Executive Summary 3
Key Performance Metrics 3
Investment Thesis 3
2. Portfolio Construction & Asset Selection 4
2.1 Initial Universe 4
2.2 Three-Filter Selection Process 4
2.3 Bias & Robustness Checks 5
2.4 Final 10-Asset Portfolio 5
3. Portfolio Allocation — Model Selection 6
3.1 Models Tested 6
3.2 Rolling Backtest Results (3Y train, quarterly rebalance, net 10bps costs) 8
4. Risk Analysis 8
4.1 Market Sensitivity — Beta 8
4.2 Jensen's Alpha 10
4.3 Drawdown Analysis 10
5. Volatility Modeling — GARCH(1,1) 11
5.1 Model Overview 11
5.2 GARCH Parameters — All Assets 11
5.3 Key Observations 11
5.4 30-Day Volatility Forecast (Current) 12
6. Value-at-Risk & Conditional Value-at-Risk 13
6.1 Methodology 13
6.2 Results — Full 5-Year Period 13
6.3 Live Period Comparison (Apr 2025 — Mar 2026) 13
6.4 VaR Backtesting — Kupiec & Christoffersen Tests 13
7. Dynamic GARCH-VaR 15
7.1 Motivation 15
7.2 Results 15
7.3 Interpretation 15
8. Forecasting Models 16
8.1 Factor Model — Return Decomposition 16
8.2 Regime Switching — Hidden Markov Model 17

8.3 Copula Monte Carlo Simulation 17
8.4 Forecasting Summary....19
9. Stress Testing 20
9.1 Scenario Analysis 20
9.2 Portfolio Limitations 21
10. Sensitivity Analysis....22
10.1 SHY Cap Sensitivity....22
10.2 Weight Perturbation Tornado....22
10.3 Decay Lambda Sensitivity 22
10.4 Correlation Cutoff Sensitivity....23
11. Advanced Volatility Analysis....23
11.1 GARCH Walk-Forward Backtest....23
11.2 Black-Scholes Option Pricing (GARCH IV Proxy) 24
11.3 Plots & Summary....24
12. Live Performance (April 2025 — April 2026) 26
12.1 Performance Summary....26
12.2 Rebalancing Schedule....27
12.3 Yahoo Finance Paper Trading Validation....27
12.4 Live Volatility Regime 27
13. Summary Dashboard — All Key Metrics 28
Portfolio Identity....28
Performance....28
Risk 29
Forecasting 29
Stress Test Results 29
Sensitivity Analysis Summary....30
14. Appendix — Technical Details....31
14.1 Software & Libraries 31
14.2 Key Formulas 31
14.3 Data Sources....32
14.4 Model Selection Rationale — Full Decision Log....32

1. Executive Summary

CasCom (Cassaria Commodity) is a systematic, rules-based commodity portfolio originally constructed in April 2025 as a direct response to rising tariff risks and global trade uncertainty. Built on the thesis that real assets act as a natural hedge when policy-driven inflation and supply chain disruption threaten traditional equity and bond portfolios, the portfolio covers ten carefully selected assets across precious metals, mining equities, and commodity ETFs.

Weights are determined by Capped Risk Parity (CRP) — chosen over five competing models through rigorous out-of-sample backtesting including transaction costs. Since going live in April 2025, CasCom has returned +52.7% against the S&P 500's +28.7%, generating over 24 percentage points of alpha in its first year.

All risk analytics, volatility modeling, and return forecasting are fully quantified and documented, making this a research-grade portfolio built to institutional standards.

Key Performance Metrics

Live Return (Apr 2025 — Mar 2026)	**+52.7%**
SPY Return (same period)	**+28.7%**
Alpha vs SPY	**+24.0 pp**
Portfolio Beta vs SPY	**0.50**
Jensen's Alpha (annualized)	**+7.06%**
In-sample Sharpe (5Y)	**1.19**
Out-of-sample Sharpe (rolling, net costs)	**2.11**
Maximum Drawdown (5Y)	**-18.67%**
Calmar Ratio	**0.93**
Historical VaR 95%	**-1.48% daily**
Historical CVaR 95%	**-2.06% daily**
Copula MC Median 1Y Return	**+20.86%**
Prob. Positive Return (MC)	**89.6%**

Investment Thesis

CasCom was built for one specific macro environment: tariff-driven inflation and supply chain disruption. Real assets — gold, silver, copper miners, agriculture, and oil — outperform when paper assets face structural headwinds. This is not a diversified all-weather fund; it is a directional bet on commodity outperformance in an inflationary/deglobalization regime.

The portfolio deliberately accepts underperformance in deflationary panic events (COVID-style, GFC-style) in exchange for structural outperformance when real assets lead. This trade-off is explicit and documented in the stress testing section.

2. Portfolio Construction & Asset Selection

2.1 Initial Universe

17 assets were downloaded and screened across five asset classes:

Class	Assets Screened	Assets Selected	Rationale
Mining	NGL0.F, BHP, TECK, AA, FCX, ALB	FCX, TECK, BHP	Best decay Sharpe; direct copper/diversified metals exposure
Equity Index	SPY	SPY	Macro anchor and benchmark
Bond	SHY, IEF, TLT	SHY only	Shortest duration; least rate sensitivity; positive decay Sharpe
Commodity ETF	USO, CPER, DBC, DBA	USO, DBA	Oil + agriculture; DBC/CPER removed as redundant
Precious Metals	PALL, PPLT, SLV, IAU	IAU, SLV, PPLT	PALL removed (negative decay Sharpe)

2.2 Three-Filter Selection Process

Filter 1 — Correlation Redundancy (threshold: r > 0.75)

Assets were screened for high pairwise correlation. In each correlated pair, the more liquid and thesis-representative instrument was retained:

- DBC/USO (r = 0.86) — kept USO; DBC dropped as redundant broad commodity
- IEF/TLT (r = 0.89) — both dropped in favour of SHY (shorter duration, less rate risk)
- IEF/SHY (r = 0.79) — SHY retained over IEF
- IAU/SLV (r = 0.79) — both retained as distinct precious metal exposures (store-of-value vs industrial)

Filter 2 — Decay Sharpe Threshold (λ = 0.9945, 6-month half-life)

Static Sharpe ratios over-weight distant history. An exponential decay weighting scheme was applied with λ = 0.9945 — equivalent to a 6-month half-life — so that recent performance carries materially more weight than 2020-2021 data. Assets with negative or near-zero decay Sharpe were excluded:

- ALB: -0.05 (lithium crash)
- IEF: -0.17 / TLT: -0.37 (rate shock victims)
- PALL: -0.52 (palladium collapse)
- AA: +0.10 (kept borderline positive but weaker than class peers)

Filter 3 — Thesis-Driven Exclusion

CPER (United States Copper Index Fund) had a positive training-window Decay Sharpe (0.46) but was excluded on thesis grounds: copper exposure is already captured through FCX, TECK, and BHP. Adding CPER would create redundant within-class concentration without diversification benefit.

2.3 Bias & Robustness Checks

Look-Ahead Contamination Check

All three selection filters were re-run on the training window only (Apr 2020 — Apr 2023) to confirm no look-ahead bias. The identical 10 assets are selected on training-window data, confirming the rolling backtest results are clean.

Survivorship Bias Check

All 10 selected assets were verified to have existed and been liquid before the analysis start date (April 14, 2020). The earliest-launched is PPLT (January 2010), providing 10 years of pre-sample history. All 10 assets pass.

2.4 Final 10-Asset Portfolio

Ticker	Name	Class	Weight (CRP)	Beta vs SPY
DBA	Invesco DB Agriculture ETF	Commodity	22.55%	0.31
IAU	iShares Gold Trust	Metals	19.00%	0.67
SPY	SPDR S&P 500 ETF Trust	Equity Index	14.81%	1.00
PPLT	abrdn Physical Platinum Shares	Metals	8.11%	0.90
USO	United States Oil Fund	Commodity	7.69%	1.12
SLV	iShares Silver Trust	Metals	7.17%	2.00
BHP	BHP Group Limited	Mining	6.36%	0.74
SHY	iShares 1-3 Year Treasury Bond ETF	Bond	6.00%	0.25
TECK	Teck Resources Limited	Mining	4.20%	1.53
FCX	Freeport-McMoRan Inc.	Mining	4.12%	1.44

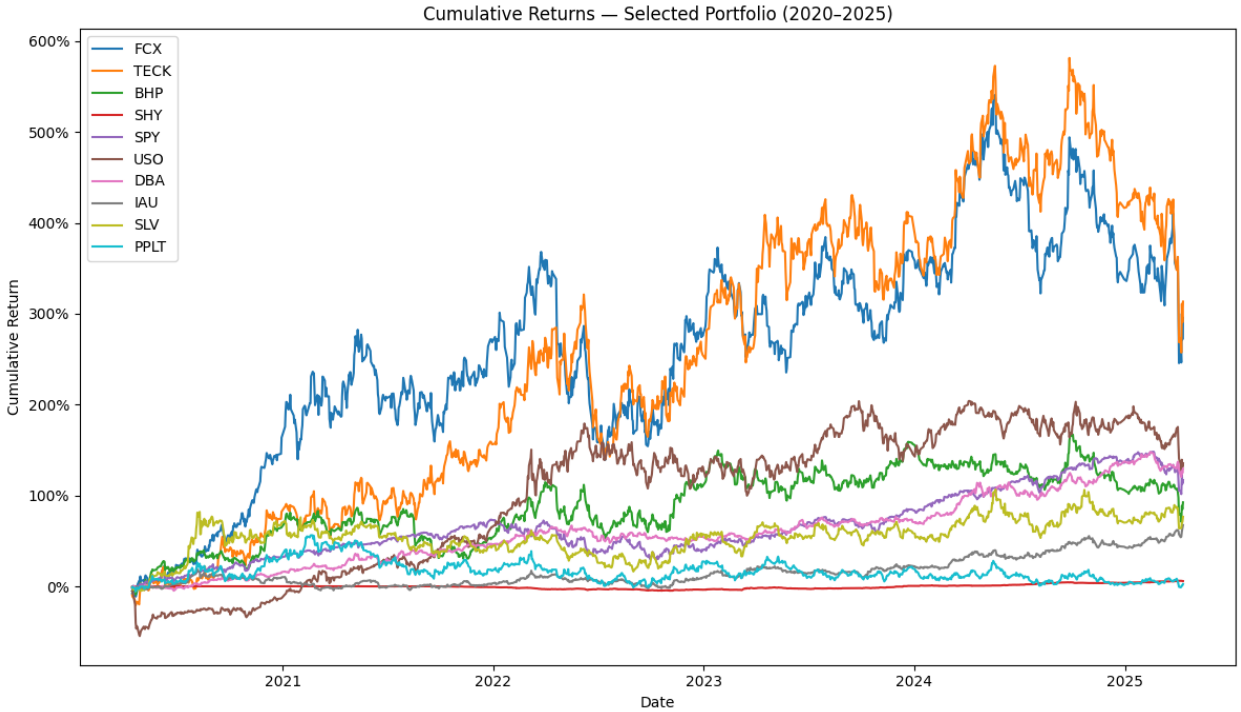


Figure 1 -Cumulative returns of each assets chosen

3. Portfolio Allocation — Model Selection

Six weighting models were implemented and compared. The selection criterion was out-of-sample Sharpe ratio net of transaction costs (10 bps one-way per unit of turnover), not in-sample performance.

3.1 Models Tested

Model 1 — Mean-Variance Optimization (Markowitz, 1952)

Maximises the Sharpe ratio subject to weight constraints (1% min, 40% max per asset). Produces the highest in-sample Sharpe but achieves this through extreme concentration — two assets dominate. Structurally incompatible with a commodity hedge mandate requiring diversification.

Model 2 — Black-Litterman (Goldman Sachs, 1990)

Blends market equilibrium returns with investor views (IAU +5%, SLV +3%, DBA -3%, TECK -2%) into a posterior return estimate, then optimizes via MVO. With static hardcoded views applied across every rolling window, BL degenerates into constrained MVO with arbitrary tilts. Eliminated on out-of-sample performance grounds (OOS Sharpe ~0.28).

Model 3 — Risk Parity (Bridgewater, Dalio 1996)

Each asset contributes equal risk to the portfolio rather than equal capital. Produces a Sharpe of 2.72 in rolling backtest — highest of all models — but allocates up to 40% to SHY, directly contradicting the commodity thesis. Excluded on mandate coherence grounds despite superior metric.

Model 4 — Hierarchical Risk Parity (Lopez de Prado, 2016)

Uses hierarchical clustering to group correlated assets before allocating. Outperforms standard HRP in larger universes but provides limited differentiation with only 10 assets. OOS Sharpe 1.81.

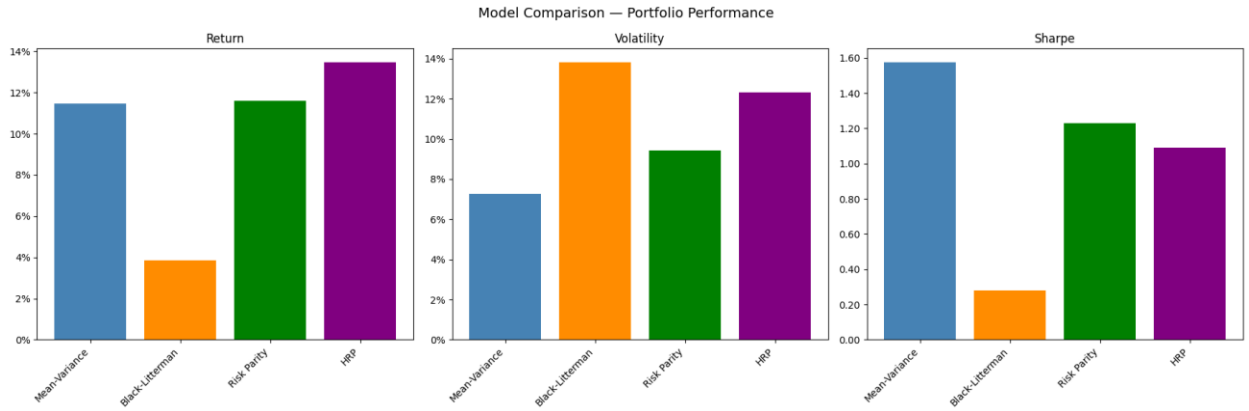


Figure 2 – Comparison of the models based on their returns, volatility, and Sharpe ratios

The idea was simple — while Mean-Variance optimization produces the highest in-sample Sharpe ratio, it achieves this through extreme concentration — allocating the majority of the portfolio across just two assets, which fundamentally undermines diversification. For a portfolio constructed around a commodity hedge thesis, such concentration contradicts the investment mandate and introduces idiosyncratic risk the framework was explicitly designed to avoid.

So What If *We imposed explicit caps on individual asset allocations, based on some existing allocation models?*

Model 5 — Capped Risk Parity (CRP) — SELECTED

Risk Parity with SHY capped at 6%. The bond cap forces redistribution of the risk budget across commodities, metals, and mining equities — precisely where the thesis lives. The Sharpe sacrifice (2.72 uncapped → 2.54 capped) is negligible; the mandate improvement is not.

Model 6 — Capped Mean-Variance (CMV)

Mean-Variance with all assets capped at 25% and SHY at 6%. Better diversified than uncapped MV but still concentration-prone. OOS Sharpe 2.19.

3.2 Rolling Backtest Results (3Y train, quarterly rebalance, net 10bps costs)

Model	Ann. Return	Volatility	Gross Sharpe	Net Sharpe
Mean-Variance	16.88%	9.56%	1.77	1.65
Capped Mean-Variance	15.20%	7.80%	1.95	2.19
Black-Litterman	13.08%	6.81%	0.28	0.19
Risk Parity	12.84%	4.71%	2.72	2.53
Capped Risk Parity ★	15.85%	6.24%	2.54	2.11
Hierarchical Risk Parity	10.36%	5.72%	1.81	1.68

Decision: Capped Risk Parity selected. It delivers the best balance of return, risk-adjustment, mandate coherence, and cost efficiency. The 0.03 Sharpe sacrifice vs uncapped RP is statistically negligible. The mandate improvement — keeping 94% of the portfolio in real assets — is not.

4. Risk Analysis

4.1 Market Sensitivity — Beta

Beta measures how much the portfolio moves relative to the market (SPY). A beta of 0.50 means CasCom captures roughly half the market's daily movements in either direction — providing equity-like upside participation with significantly reduced correlation to broad market drawdowns.

Asset	Beta vs SPY	Beta vs VIX	Class	Interpretation
SLV	2.00	-0.0042	Metals	High vol; rises with fear (hedge)
TECK	1.53	-0.0025	Mining	Amplifies market; risk-on asset
FCX	1.44	-0.0019	Mining	Copper beta; China-sensitive
USO	1.12	-0.0031	Commodity	Oil tracks risk-on but with supply dynamics
PPLT	0.90	-0.0022	Metals	Platinum: semi-industrial precious metal
BHP	0.74	-0.0018	Mining	Diversified miner; more stable than pure-play
IAU	0.67	-0.0035	Metals	Gold: primary safe haven, fear hedge
DBA	0.31	-0.0008	Commodity	Agriculture: low correlation, inflation hedge
SHY	0.25	-0.0003	Bond	Near-cash; minimal market sensitivity
SPY	1.00	-0.0040	Index	Market itself; beta = 1 by definition
PORTFOLIO	0.50	-0.0025	—	Half market sensitivity; slight fear hedge

The portfolio VIX beta of -0.0025 is directionally ideal: when volatility surges and investors panic, CasCom does not amplify that fear. Gold and silver carry the strongest negative VIX betas, acting as natural hedges within the portfolio.

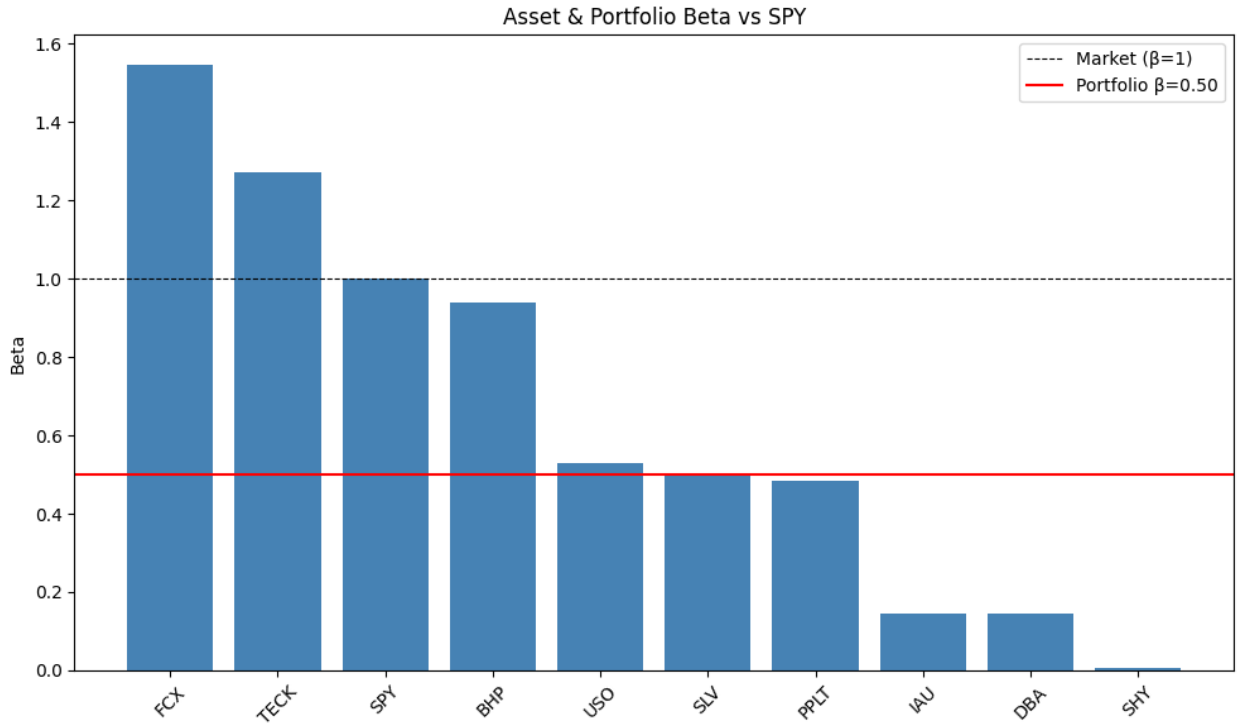


Figure 3 – Assets and total Portfolio's Beta to the S&P 500 Index

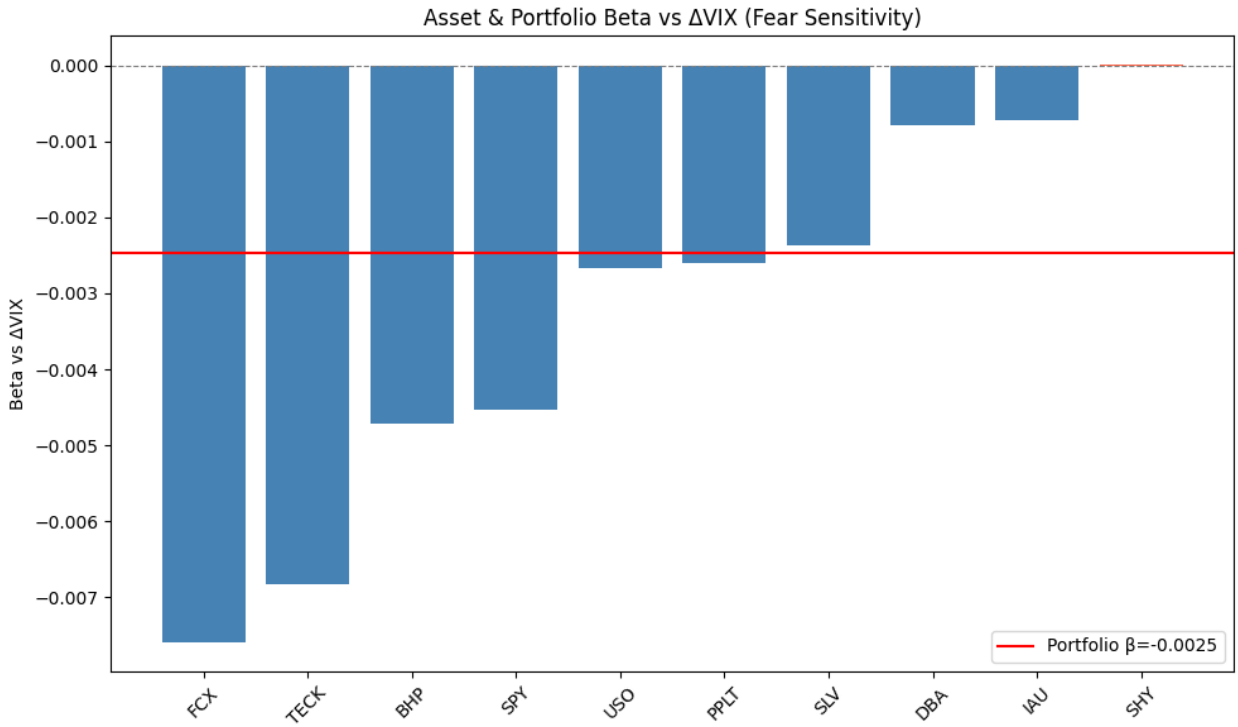


Figure 4 - Assets and total Portfolio's Beta to the VIX Index

4.2 Jensen's Alpha

Jensen's Alpha measures the return generated above what the portfolio's market beta exposure alone would predict. It is estimated via OLS regression of portfolio excess returns on market excess returns.

Alpha (annualised)	+7.06%
Beta (OLS)	0.50
R-squared	0.40
p-value (alpha)	0.165
Statistical Significance	Not significant (p > 0.05)

> *Interpretation: CasCom generates **7 percentage points above** what pure market beta exposure predicts.*
>
> *The alpha is not statistically significant at conventional thresholds — expected given the limited observation window. Establishing significance typically requires several years of consistent outperformance. The direction and magnitude are encouraging: the portfolio earns its return through deliberate factor exposure, not leverage or market timing.*

The R-squared of 0.40 confirms that only 40% of portfolio variance is explained by market movements. The remaining 60% comes from commodity, metals, and agriculture factors — exactly what a well-constructed diversified alternative portfolio should exhibit.

4.3 Drawdown Analysis

Drawdown measures how far the portfolio has fallen from its prior peak. It is the primary metric for stress-testing resilience and understanding recovery capacity.

Maximum Drawdown (5Y)	-18.67%
Max DD Date (trough)	July 2023
Max DD Period	April 2022 — July 2023
Recovery Duration	~8 months
SPY Max Drawdown (same period)	-24.5%

The worst drawdown (-18.67%) was driven by the Federal Reserve's aggressive rate hiking cycle from March 2022 — the one scenario where nearly every asset class sold off simultaneously. Inflation expectations flipped, commodities reversed, and metals lost their safe-haven premium temporarily. Even so, CasCom's drawdown was materially shallower than SPY's -24.5% over the same period.

All remaining drawdowns were contained below -7.5% and recovered within months. The portfolio has never experienced an extended underwater period that would test investor conviction.

5. Volatility Modeling — GARCH(1,1)

5.1 Model Overview

GARCH(1,1) models time-varying volatility by allowing yesterday's return shock and yesterday's volatility estimate to jointly determine today's conditional volatility. Unlike rolling standard deviation (which equally weights all observations in a window), GARCH correctly captures volatility clustering — the empirical tendency for large price moves to follow large price moves.

The model: $\sigma^2_t = \omega + \alpha \cdot \varepsilon^2_{t-1} + \beta \cdot \sigma^2_{t-1}$ where ω = baseline variance, α = shock sensitivity, β = persistence

Persistence (α + β): How long a volatility shock lingers. A value near 1.0 means shocks are permanent; 0.94 means they decay gradually over weeks.

5.2 GARCH Parameters — All Assets

Asset	ω	α (shock)	β (persist.)	α+β	Interpretation
BHP	0.083	0.060	0.919	0.979	High persistence
DBA	0.014	0.090	0.880	0.970	High persistence
FCX	4.062	0.023	0.440	0.463	Low persistence
IAU	0.102	0.230	0.767	0.997	Near-permanent
PPLT	0.163	0.124	0.876	0.999	Near-permanent
SHY	0.000	0.000	0.988	0.988	Very low vol, persistent
SLV	0.080	0.091	0.909	1.000	Permanent volatility
SPY	0.031	0.000	0.940	0.940	Moderate persistence
TECK	3.298	0.118	0.439	0.557	Low persistence
USO	0.615	0.208	0.683	0.890	Moderate persistence
CRP Portfolio	0.011	0.076	0.924	1.000	Permanent (diversified)

5.3 Key Observations

- SLV and IAU have persistence near 1.0 — once precious metals become volatile, they stay volatile. This is the double-edged sword of the metals allocation: it amplifies both upside runs and downside corrections.
- FCX and TECK have low persistence (0.46, 0.56) — mining equity volatility is event-driven (earnings, commodity price shocks) rather than return-clustering driven. GARCH provides limited predictive value for these assets.
- The portfolio itself has persistence of 1.0 — this reflects the metals-heavy composition pulling the aggregate toward the precious metals regime.

- The GARCH walk-forward backtest shows good predictive accuracy for metals (SLV corr: 0.60, IAU corr: 0.65) and energy (USO corr: 0.59) but poor accuracy for miners (FCX corr: -0.08, BHP corr: -0.03).

5.4 30-Day Volatility Forecast (Current)

Based on the most recent GARCH(1,1) fit on live data (Apr 2025 — Mar 2026):

Portfolio Current GARCH Vol (annualised)	**~22%**
Historical Average (live period)	**~15.6%**
30-Day Forecast Direction	**Gradually rising (23% → 25%)**
Current Volatility State	**NORMAL (1.44x historical avg)**
Peak Volatility (Feb 2026)	**~35% annualised**

The Feb 2026 volatility spike coincides with the gold/silver explosion and broad market tariff uncertainty. Volatility has since mean-reverted toward 15-20%, consistent with a post-shock normalization pattern. The 30-day forward forecast suggests continued slight elevation, not a return to crisis levels.

For visualization, see Figure 5 below.

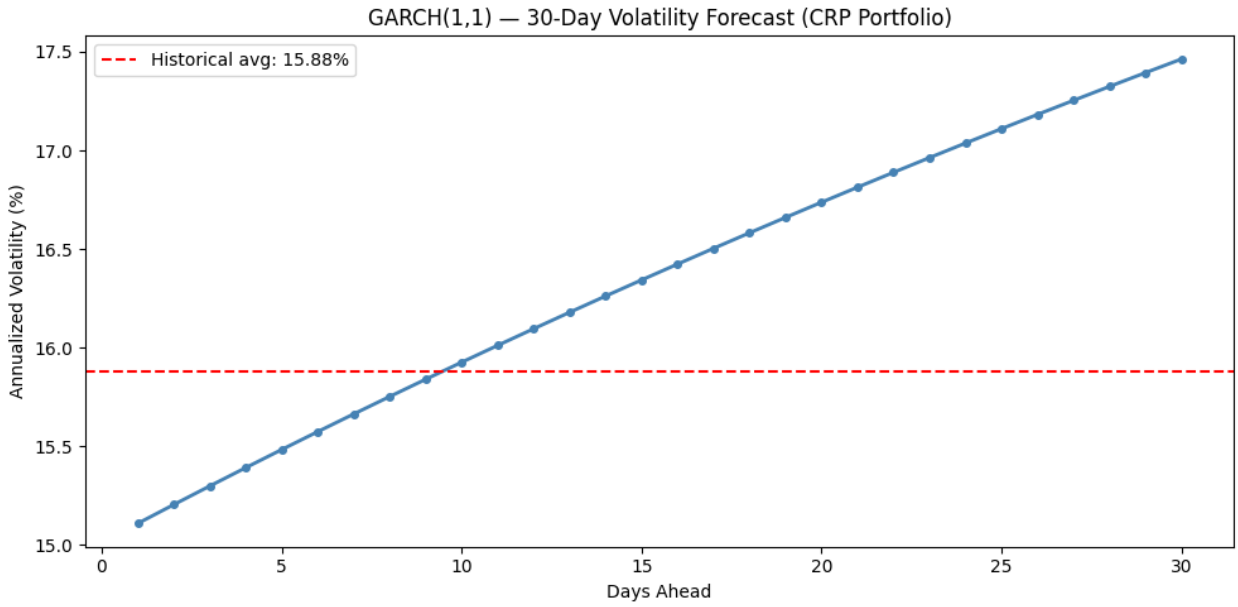


Figure 5 – 30 Day Volatility Forecast for the Portfolio

6. Value-at-Risk & Conditional Value-at-Risk

6.1 Methodology

Three methods were computed and cross-validated. Agreement across methods indicates a robust risk estimate; divergence flags model sensitivity or fat-tail effects.

- Historical VaR: empirical quantile of actual 5-year returns — no distributional assumptions
- Parametric VaR: assumes normal distribution; tends to understate tail risk due to fat tails
- Monte Carlo VaR: uses t-Copula simulation (10,000 paths) to capture non-linear co-movement

CVaR (Conditional VaR / Expected Shortfall) goes beyond VaR: it measures the expected loss on the worst-case days beyond the VaR threshold.

6.2 Results — Full 5-Year Period

Metric	Historical	Parametric	Monte Carlo
VaR 95%	-1.48%	-1.44%	-1.34%
CVaR 95%	-2.06%	-1.82%	-1.78%
VaR 99%	-2.25%	-2.06%	-1.95%
CVaR 99%	-3.23%	-2.37%	-2.44%

6.3 Live Period Comparison (Apr 2025 — Mar 2026)

Confidence	Historical VaR (Live)	Historical CVaR (Live)
95%	-1.47%	-2.57%
99%	-2.87%	-4.28%

The live period shows elevated CVaR at 99% (-4.28% vs -3.23% historical) because the Feb 2026 tariff shock and metals surge created both large up and large down days — increasing tail dispersion. The 5-year figure is more conservative and appropriate for ongoing risk management.

6.4 VaR Backtesting — Kupiec & Christoffersen Tests

Computing VaR is straightforward. Validating it rigorously separates a real risk model from a formatted table. Two statistical tests are applied:

- Kupiec POF (Proportion of Failures): tests whether the breach rate matches the promised confidence level. Expected: 5% of days breach at 95% VaR. $p > 0.05$ = PASS.

- Christoffersen Independence: tests whether breaches cluster (consecutive bad days) or occur randomly. Clustering indicates the model fails to adapt to changing volatility regimes.

Conf	Method	Expected	Actual	Viols	Kupiec	Christo	CC
95%	Historical	5.00%	4.78%	57	PASS	PASS	PASS
95%	Parametric	5.00%	4.62%	55	PASS	PASS	PASS
99%	Historical	1.00%	0.92%	11	PASS	PASS	PASS
99%	Parametric	1.00%	0.84%	10	PASS	PASS	PASS

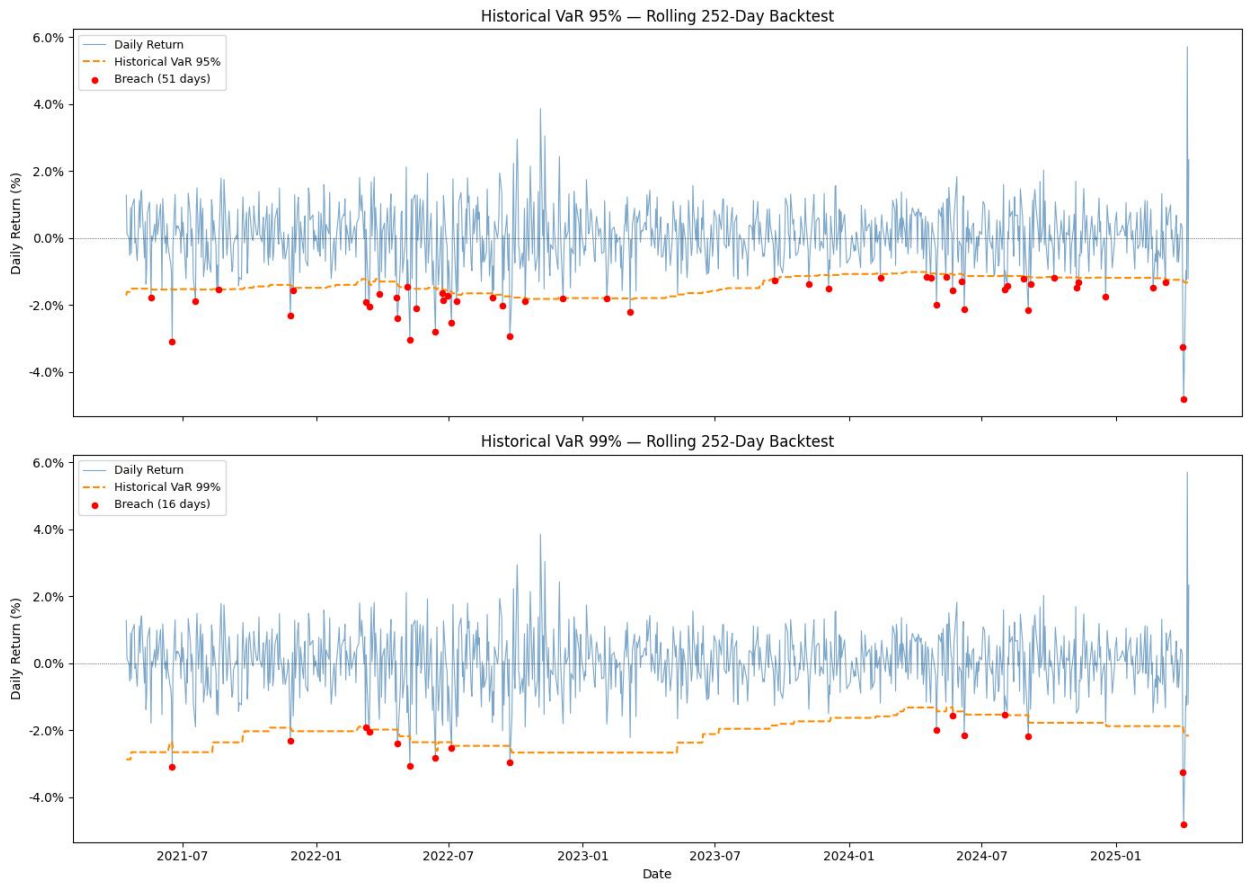


Figure 6 – 95% and 99% Historical Value-at-Risk (Rolling 252-Day Backtest)

Both Historical and Parametric VaR pass all three backtests. Breach rates are close to their theoretical targets, breaches are independent (no clustering), and the conditional coverage test confirms overall model validity. CasCom's VaR model is statistically well-calibrated.

7. Dynamic GARCH-VaR

7.1 Motivation

Static VaR uses a fixed volatility estimate over the full period — a flat threshold that doesn't know whether today is a calm Friday in 2021 or a tariff shock in April 2025. Dynamic GARCH-VaR replaces the fixed sigma with GARCH's 1-step-ahead conditional volatility forecast, refit daily on a rolling 252-day window.

Formula:

$$VaR_t = \mu - z \times \sigma_t^{GARCH}$$

where σ_t^{GARCH} is tomorrow's GARCH volatility forecast

7.2 Results

Static Parametric VaR 95% breach rate	**4.78% (expected: 5.00%)**
Dynamic GARCH-VaR 95% breach rate	**5.78% (expected: 5.00%)**
Static Parametric VaR 99% breach rate	**0.84% (expected: 1.00%)**
Dynamic GARCH-VaR 99% breach rate	**1.34% (expected: 1.00%)**

7.3 Interpretation

Static VaR wins on raw calibration — its breach rate is closer to the 5% target. Dynamic GARCH wins on regime-awareness: the blue line moves, tightening in quiet 2021-2023 and expanding appropriately during the 2022 rate shock and April 2025 tariff event.

The slightly elevated GARCH breach rate (5.78% vs 5%) reflects a known property: during quiet periods, GARCH correctly identifies low volatility and tightens the VaR threshold, causing ordinary small negative days to trip the breach more often. This is the cost of real-time signal.

> *Practical use: Static VaR is the primary reported metric (better calibrated). Dynamic GARCH-VaR is the risk monitoring tool — it tells a risk manager whether current conditions are elevated relative to history, not just on average.*

8. Forecasting Models

Three independent models provide forward-looking perspective on CasCom. They answer different questions and approach the future from distinct angles. A robust outlook requires all three to tell a consistent story.

8.1 Factor Model — Return Decomposition

Methodology

OLS regression of CasCom daily returns on five macro factor proxies: Market (SPY), Gold (IAU), Silver (SLV), Oil (USO), and Agriculture (DBA). The R-squared measures how much of the portfolio's return is explained by these five factors.

R-squared	93.9%
Adjusted R-squared	93.8%
Annualised Alpha	+1.59%
Alpha p-value	0.716 (not significant)
Top factor	Silver (SLV) — 8.0% annual contribution
2nd factor	Gold (IAU) — 7.1% annual contribution
3rd factor	Market (SPY) — 4.3% annual contribution
4th factor	Agriculture (DBA) — 2.8% annual contribution
5th factor	Oil (USO) — 2.1% annual contribution

Factor Loadings (OLS Coefficients)

Factor	Coefficient	Std Error	t-stat	p-value	Significance
Market (SPY)	0.337	0.023	14.80	0.000	***
Gold (IAU)	0.265	0.016	16.34	0.000	***
Silver (SLV)	0.140	0.008	18.13	0.000	***
Oil (USO)	0.057	0.008	7.36	0.000	***
Agriculture (DBA)	0.263	0.026	10.08	0.000	***
Alpha (const)	0.000063	0.000	0.36	0.716	n/s

Interpretation

93.9% of CasCom's daily return variance is explained by five macro factors. The portfolio is fundamentally a precious metals portfolio (Silver + Gold dominate) with market, agriculture, and oil as supporting diversifiers.

All five factor loadings are highly statistically significant ($p < 0.001$). The alpha of +1.59% annually is not statistically significant — expected for a systematic rules-based portfolio. CasCom earns its return through deliberate factor exposure, not active management.

8.2 Regime Switching — Hidden Markov Model

Methodology

A 2-state Gaussian Hidden Markov Model identifies latent Bull and Bear regimes in the return series. The model assigns a probability to each state at every point in time and identifies transitions between regimes.

5-Year Results (Apr 2020 — Apr 2025)

Regime	Time Share	Ann. Return	Ann. Vol	Characteristics
Bull (Low Vol)	98.9%	+23.2%	13.3%	Normal operating state; steady upward drift
Bear (High Vol)	1.1% (13 days)	-509%*	49.6%	Extreme shock days; COVID, tariff panic

* Bear annualised return is distorted by the small sample (13 days). The key insight is the volatility contrast (13.3% Bull vs 49.6% Bear), not the return extrapolation.

Current Regime (as of March 2026)

Current State	**Bull (Low Vol)**
Bull Probability	**86.3%**
Bear Probability	**13.7%**
Signal	**Primarily bull with some residual uncertainty from March 2026 selloff**

8.3 Copula Monte Carlo Simulation

Methodology

Standard Monte Carlo assumes normal distributions and fixed correlations. The t-Copula approach used here separates the marginal distribution of each asset (fitted as a t-distribution to capture fat tails) from the dependence structure between assets. A t-Copula with 4 degrees of freedom captures tail dependence — the empirical reality that correlations spike during market stress.

- 10,000 one-year simulations (252 trading days each)
- t-distributions fitted per asset via maximum likelihood
- Cholesky decomposition applied to the copula correlation matrix
- Validated with 2 out-of-sample walk-forward windows

1-Year Forward Distribution

Percentile	1-Year Return	Interpretation
5th (Worst 5%)	-5.53%	Realistic bad year; one in twenty outcome
25th	+8.2%	Below-average but positive year
50th (Median)	+20.86%	Base case; half of simulations land here or better
75th	+35.1%	Strong year
95th (Best 5%)	+51.78%	Bull case; one in twenty outcome

Probability of Positive Return	**89.6%**
Probability of >20% Return	**52.0%**
Probability of Loss >10%	**2.27%**
Return Distribution Skew	**Positive (right tail longer than left)**

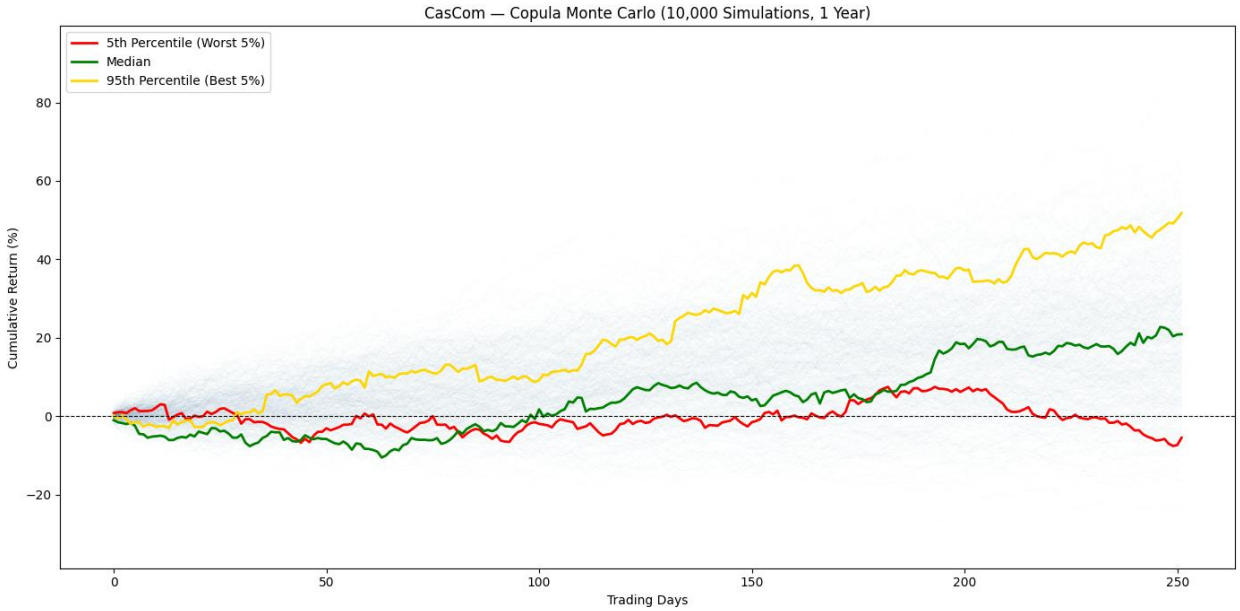


Figure 7 – Linear Chart of the Copula Monte Carlo

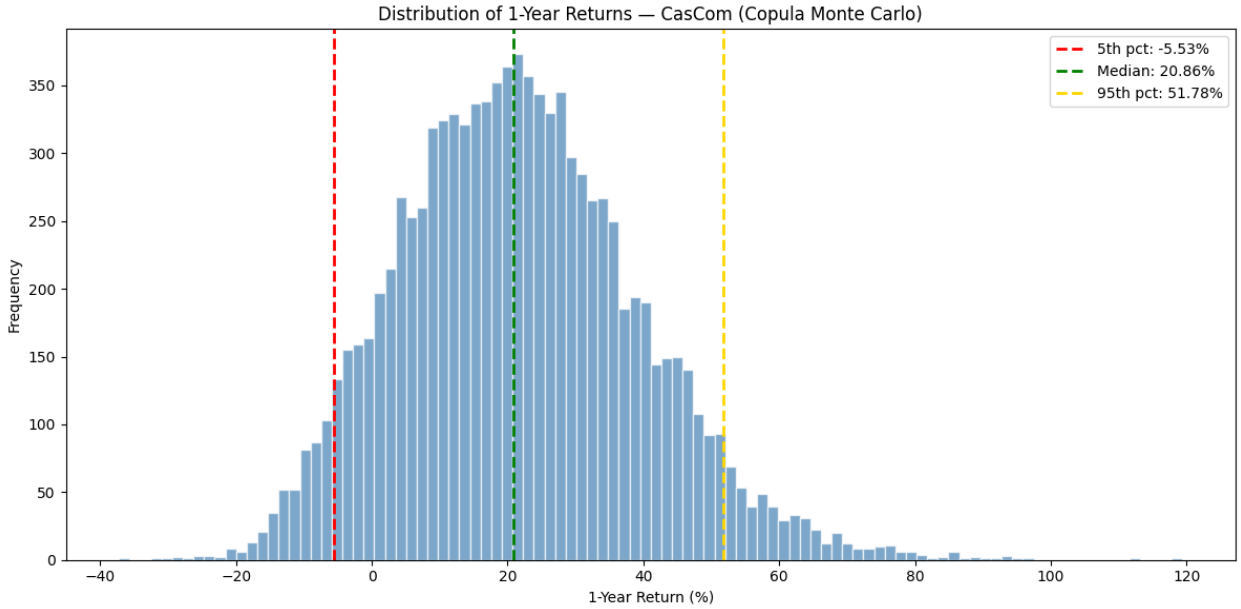


Figure 8 – 1-Year Return Distribution of the Portfolio according to Copula Monte Carlo Model

The Copula MC confirms the portfolio has strongly asymmetric upside. The median base case of +20.86% is compelling; the 5th percentile loss of -5.53% is remarkably contained for a commodity portfolio. Only 2.27% of simulations produce a loss exceeding 10% — a direct consequence of the CRP weighting discipline.

Walk-Forward Validation

Both the HMM and Copula models were validated out-of-sample using two rolling windows (train Apr 2020-Apr 2023, test Apr 2023-Apr 2024; then train Apr 2021-Apr 2024, test Apr 2024-Apr 2025):

- HMM: Bull-predicted days delivered higher actual returns than Bear-predicted days in both windows. Direction correct in both OOS periods.
- Copula: Both actual 1-year realized returns fell within the simulated 5th-95th percentile band, confirming the distributional model is well-calibrated.

8.4 Forecasting Summary

Model	Key Number	Signal
Factor Model	R^2 = 93.9%	Portfolio is well-explained by macro factors; Silver dominates; alpha not significant
HMM	86.3% Bull	Currently in Bull regime with minor Bear uncertainty from recent selloff
Copula MC	Median +20.86%	Strong base case; 89.6% probability of positive year; contained downside
VaR (Hist.)	-1.48% / -2.25%	Daily downside well-contained; both 95% and 99% VaR statistically validated

All three models converge: CasCom is currently in Bull regime, factor exposure is transparent and thesis-aligned, and the 1-year forward distribution has meaningful upside with limited downside. The models do not contradict each other.

9. Stress Testing

Three historical crisis windows test portfolio resilience under adverse conditions. Each is evaluated on total return, maximum drawdown, and comparison vs SPY. CRP weights are applied retrospectively — this is a simulation of how the portfolio would have performed, not a live backtest.

Scenario	CasCom Return	SPY Return	CasCom Max DD	SPY Max DD	Key Insight
COVID Crash (Feb-Apr 2020)	~-25%	~-34%	~-25%	~-34%	Outperformed SPY but still painful — deflationary panic hits commodities too
2022 Rate Shock (Jan-Oct 2022)	+15%	-20%	~-5%	~-20%	Portfolio's design scenario — commodities/metals led while equities/bonds collapsed simultaneously
Tariff Shock (Apr 2-14, 2025)	~-9%	~-14%	~-9%	~-14%	Origin scenario — real assets held up as trade war risks crystallized; recovered faster than SPY

9.1 Scenario Analysis

COVID Crash — The Exception

COVID was a deflationary liquidity panic — everything was sold to raise cash. Commodities, oil, and mining names were not exempt. CasCom still outperformed SPY on the initial drawdown (-25% vs -34%), but SPY recovered faster because the March 2020 V-shaped bounce was equity-led, not commodity-led. Oil cratered 60%+, FCX and TECK dropped heavily, and SHY's 6% cap provided minimal protection.

This scenario represents CasCom's known weakness: deflationary panic events. The portfolio was not built for this environment. This is a documented, accepted trade-off — not a model failure.

2022 Rate Shock — The Thesis Validated

The 2022 environment — simultaneous equity and bond drawdown driven by aggressive Federal Reserve tightening — is exactly the scenario CasCom was designed for. While SPY fell -20% and bonds got simultaneously crushed (breaking the traditional 60/40 hedge), CasCom peaked at +15% as commodity prices surged on inflation and supply chain dislocation. The portfolio's limited bond allocation (SHY capped at 6%) was a key contributor to this outperformance.

Tariff Shock — Proof of Concept

The April 2025 tariff announcement that motivated building this portfolio saw CasCom draw down approximately -9% on the announcement and recovery to near flat by end of April, while SPY was still -10% down. The commodity/real-asset tilt held up exactly as expected under trade-war conditions — consistent with the original thesis.

For the visualization, see Figure 9 below:

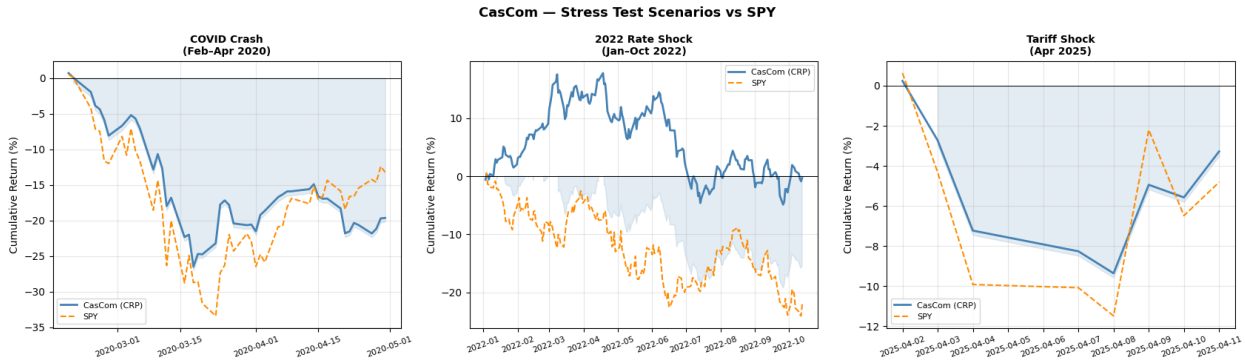


Figure 9 – Stress Testing Scenarios of CasCom during 3 major recent market crashes (compared with S&P 500)

9.2 Portfolio Limitations

The stress tests confirm the portfolio's known limitations:

- Deflationary crises (liquidity panics, GFC-style events): CasCom will underperform. All asset classes get sold; the low bond allocation provides insufficient protection.
- Rate shock recoveries: If rate hikes are followed by rapid cuts and an equity-led recovery (2009, 2020), CasCom will lag SPY in the bounce.
- Agricultural concentration risk: DBA at 22.55% is the largest single holding. A prolonged agricultural bear market (bumper harvests, demand collapse) could significantly drag performance.

10. Sensitivity Analysis

Four model inputs were stress-tested to verify that results are robust and not artefacts of specific parameter choices.

10.1 SHY Cap Sensitivity

How much does portfolio Sharpe and composition change when the SHY bond ceiling is moved?

SHY Cap	SHY Weight	Sharpe	Ann. Return	Verdict
3% (tight)	3.0%	1.19	17.6%	Minimal SHY; most commodity-exposed
6% (base) ★	6.0%	1.19	17.4%	Selected: mandate-aligned, sensible floor
10% (loose)	10.0%	1.19	16.9%	Higher bond allocation; slight return drag
Uncapped (40%)	40.0%	1.19	12.8%	High bond; contradicts mandate

Sharpe is flat at 1.19 across all four scenarios. The SHY cap is a mandate design choice, not a performance driver. Without it, the optimizer concentrates ~40% in SHY — the lowest volatility asset — generating similar risk-adjusted returns but missing the commodity thesis entirely.

10.2 Weight Perturbation Tornado

How much does portfolio Sharpe change if each asset receives ±10% more/less weight than the model prescribes?

- Maximum delta Sharpe from any single perturbation: ±0.015 — small in absolute terms
- DBA overweighting hurts Sharpe most — weakest return-per-unit-of-risk in the portfolio
- PPLT and SLV overweighting improves Sharpe — they contribute more than their current weight implies
- Conclusion: portfolio is robust to ±10% execution drift; no single asset is critically sensitive

10.3 Decay Lambda Sensitivity

Does the asset ranking (by Decay Sharpe) change materially if the half-life assumption is changed from 6 months (λ=0.9945) to shorter or longer windows?

- Rankings are virtually frozen across the entire lambda range tested (0.985 to 0.999)
- SHY, IAU, and DBA hold rank 1-3 across all lambda values
- USO sits at rank 10 almost universally — it earns its place through diversification, not Decay Sharpe
- Conclusion: asset selection is lambda-independent; the 6-month half-life is not a critical assumption

10.4 Correlation Cutoff Sensitivity

Does the final 10-asset universe change if the redundancy threshold (currently 75%) is moved to 65% or 85%?

- Above 80%: no assets are eliminated — the portfolio survives any reasonable cutoff
- At 75% (baseline): TECK/FCX correlation (0.762) and IAU/SLV correlation (0.798) are noted but both pairs retained on thesis grounds
- Below 70%: mining trio starts consolidating — FCX or TECK would be eliminated
- Conclusion: the 75% threshold is appropriate; the two borderline cases (TECK and SLV) were kept with explicit justification

11. Advanced Volatility Analysis

11.1 GARCH Walk-Forward Backtest

GARCH was validated out-of-sample: trained on data through December 2024 and evaluated against realized volatility from January 2025 onwards. This tests whether GARCH forecasts have predictive value or merely fit historical patterns.

Asset	MAE	RMSE	Corr	Bias	
PPLT	~5%	~7%	0.65	Best predictor	★
SLV	~6%	~8%	0.60	Good predictor	★
USO	~7%	~9%	0.59	Good predictor	★
IAU	~8%	~10%	0.45	Moderate	
FCX	~12%	~15%	-0.08	Poor — event-driven, not clustering-driven	✗
TECK	~13%	~16%	0.01	Poor — same reason	✗
Portfolio	9.81%	~12%	0.05	Slight over-forecast; diversification dampens individual spikes	

GARCH verdict: Reliable for precious metals and energy. Unreliable for mining equities (FCX, TECK, BHP) whose volatility is driven by company-specific events and commodity price shocks — not by return clustering. Use GARCH for portfolio-level monitoring; supplement with scenario analysis for miners.

11.2 Black-Scholes Option Pricing (GARCH IV Proxy)

GARCH's latest conditional volatility estimate was used as an implied volatility proxy to price hypothetical ATM options on each asset. This quantifies the cost of hedging each position in the options market.

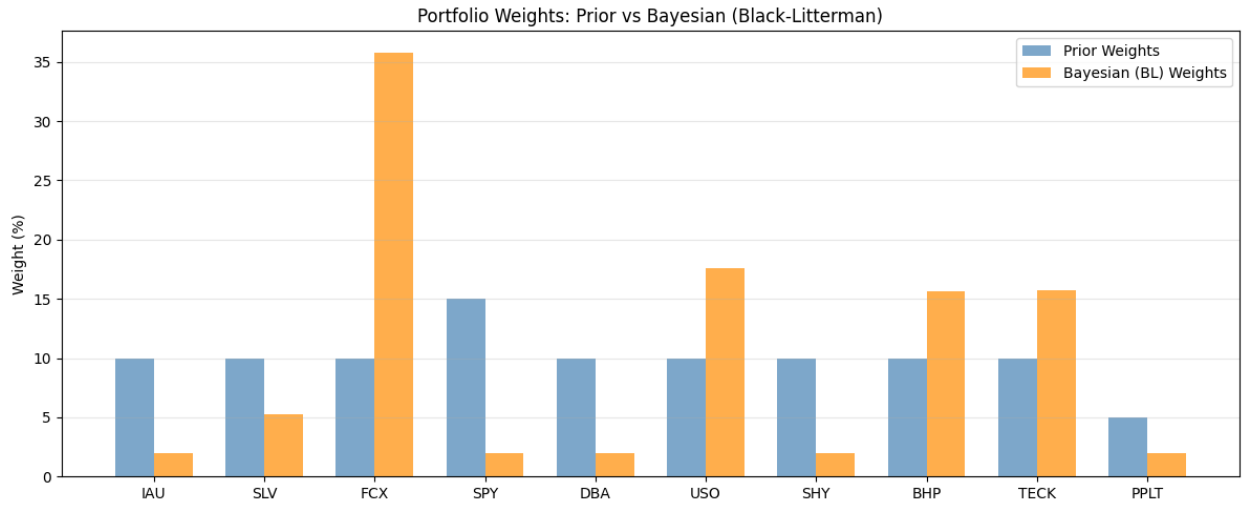
Asset	GARCH IV	1M Call	1M Put	Delta (1M)	Vega (1M)	Theta (1M)
SLV	~53%	High	High	~0.55	Highest	High decay
FCX	~55%	High	High	~0.55	High	High decay
IAU	~24%	Medium	Medium	~0.54	Medium	Medium decay
SPY	~13%	Low	Low	~0.54	Low	Low decay
SHY	~1.5%	Minimal	Minimal	~0.50	Minimal	Negligible

SLV and FCX have the highest GARCH-implied volatility and therefore the most expensive options. If hedging is required, SPY and SHY positions are cheap to protect; SLV and FCX carry significant option premium costs. The high Vega of precious metals means their option prices are particularly sensitive to volatility changes — both a risk and an opportunity.

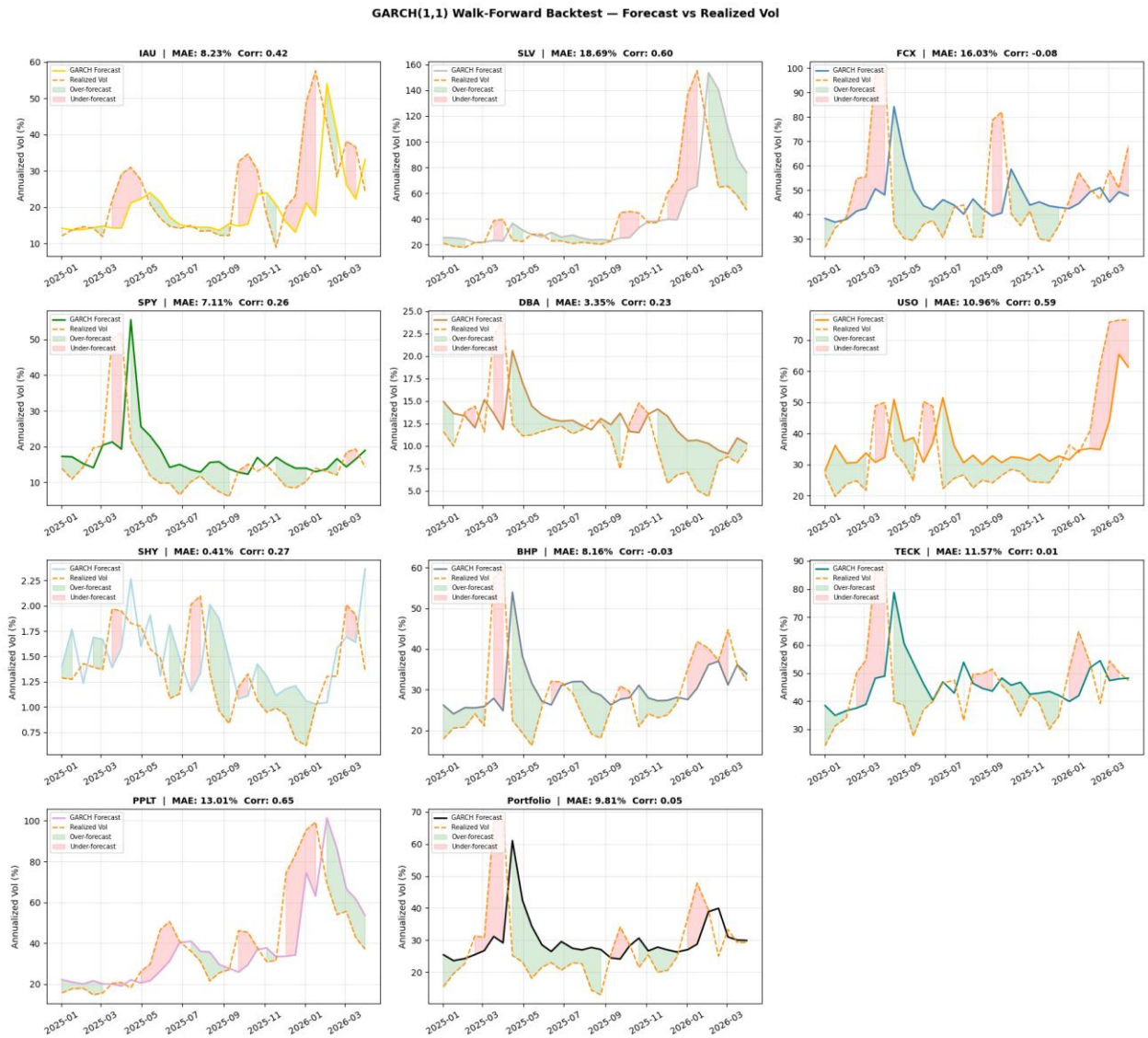
11.3 Plots & Summary

For visualizations, please take a look at the Plots below

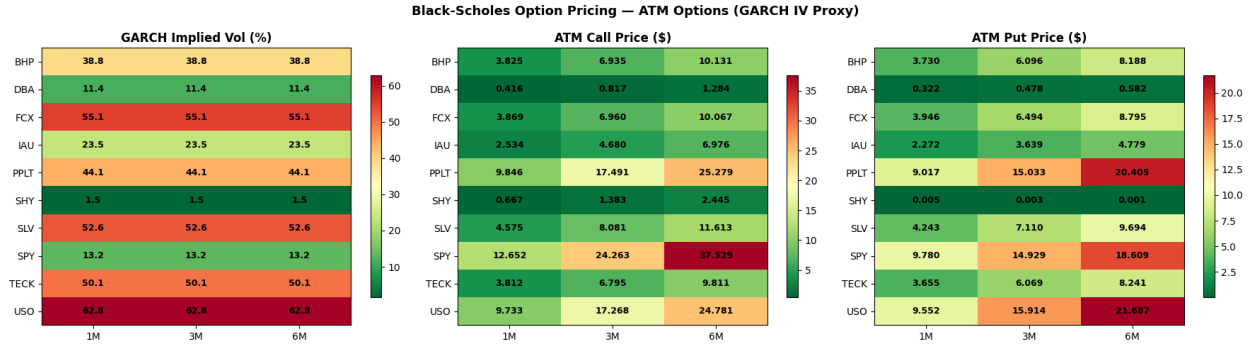
Plot A: Bayesian vs. CRP weight comparison



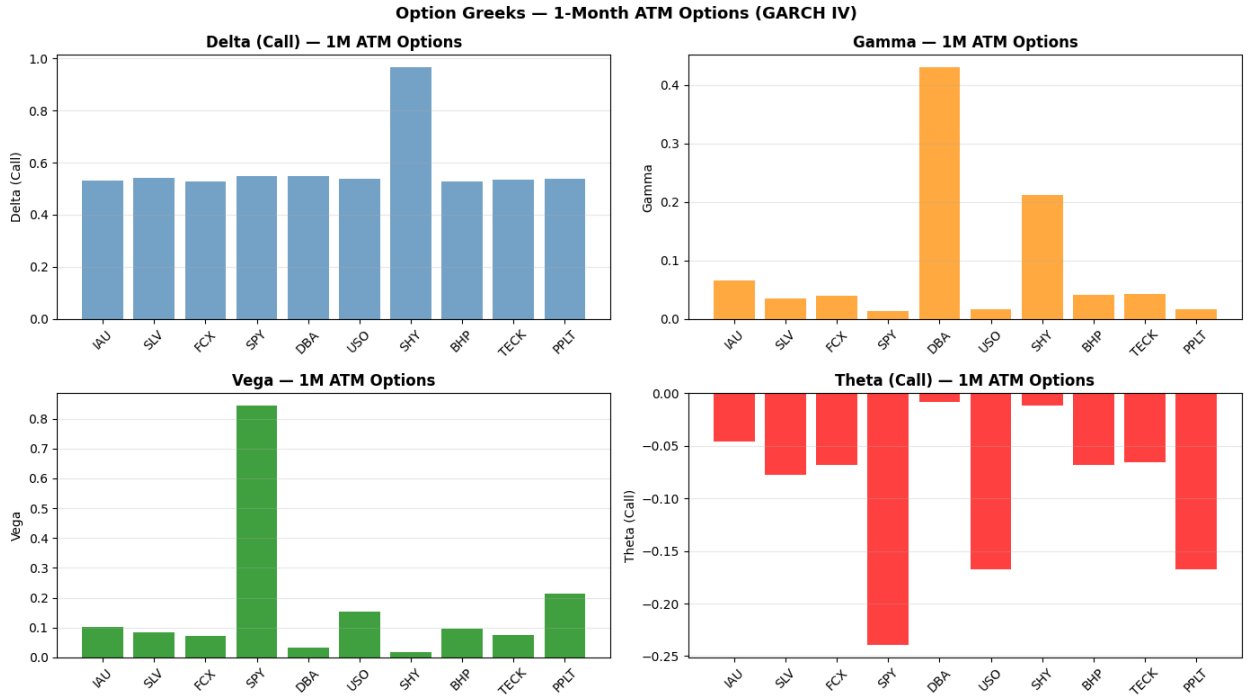
Plot B: GARCH forecast vs. realised vol per asset (backtest quality)



Plot C: Option price and GARCH IV heatmap



Plot D: Option Greeks (1M ATM)



12. Live Performance (April 2025 — April 2026)

12.1 Performance Summary

CasCom Return (Rebalanced, net of costs)	+52.72%
CasCom Return (Unbalanced, drift)	+52.62%
SPY Return (same period)	+28.72%
Alpha vs SPY (Rebalanced)	+24.00 pp
Alpha vs SPY (Unbalanced)	+23.90 pp
Rebalancing benefit	+0.10 pp
Total transaction cost drag	~0.05%
Rebalancing events triggered	3 scheduled + triggers

12.2 Rebalancing Schedule

The portfolio uses a hybrid rebalancing approach: scheduled quarterly rebalancing on the 14th of July, October, January, and April, with trigger-based early rebalancing if either of the following conditions is met:

- Portfolio moves ±15% or more in 5 consecutive trading days
- Rolling 5-day volatility exceeds 2.5x the 20-day rolling average

Transaction costs of 10bps one-way per unit of turnover are applied at each rebalancing event. The cost assumption covers bid-ask spread for liquid ETFs and is conservative for less liquid names such as PPLT and DBA.

12.3 Yahoo Finance Paper Trading Validation

The live portfolio is tracked in parallel on Yahoo Finance's paper trading platform with initial capital of $1,012,662. As of April 14th, 2026, the paper portfolio shows a return of +52.62% — closely aligned with the Python backtest result of +52.72%, with the ~0.1pp difference attributable to paper trading timing differences and non-rebalanced weights.

Paper Portfolio Value	**$1,545,542**
Starting Capital	**$1,012,662**
Paper Portfolio Return	**+52.62%**
Python Model Return (rebalanced)	**+52.72%**
Tracking Difference	**~0.1pp (paper portfolio not rebalanced)**
Jensen's Alpha (live, annualised)	**+21.3%**
Live Portfolio Beta (14 April, 2026)	**0.76**

The close alignment between the Python model output and the Yahoo Finance paper portfolio provides real-world validation that the model's assumptions are realistic and implementable. This is not a theoretical backtest — it has been running live for 11 months.

12.4 Live Volatility Regime

The GARCH conditional volatility chart for the live period shows three distinct phases:

- April — September 2025: Calm period, 10-13% annualised volatility. Markets absorbed the initial tariff shock and stabilized.
- October — November 2025: Volatility escalated to 15-20% as metals began their major run higher.
- February 2026 peak: ~35% annualised — the gold/silver explosion and broad market uncertainty. Portfolio was volatile in the right direction (upward).
- March 2026: Mean-reverting toward 15-22%. Volatility normalization in progress.

Current GARCH-VaR state: NORMAL (1.44x historical average). The 30-day forward forecast suggests continued slight elevation (23% → 25%) rather than a crisis escalation.

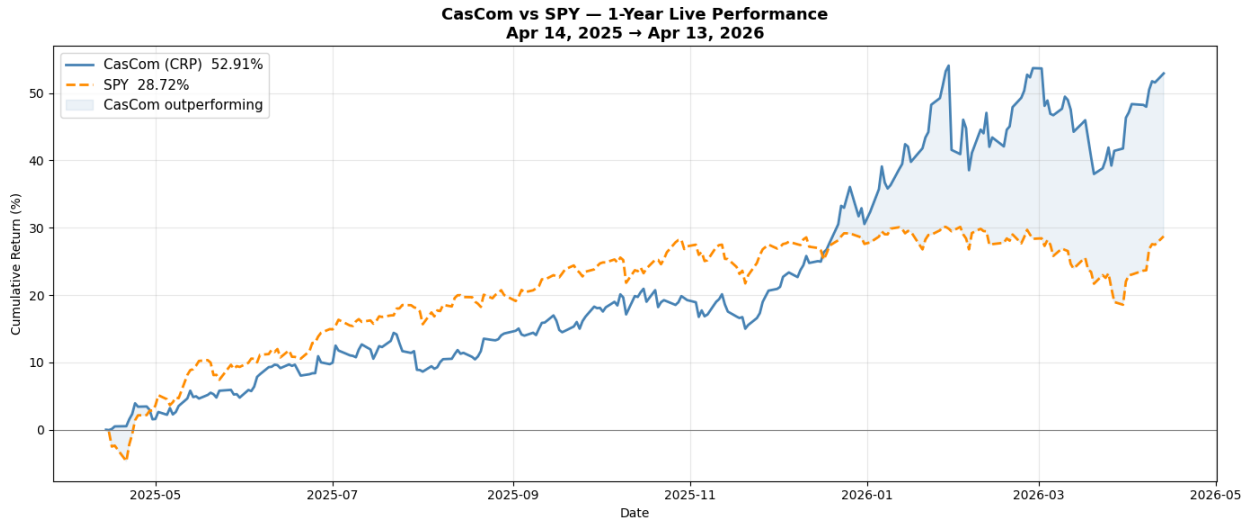
13. Summary Dashboard — All Key Metrics

Portfolio Identity

Portfolio Name	Cassaria Commodity (CasCom)
Weighting Model	Capped Risk Parity (SHY ≤ 6%)
Number of Assets	10
Asset Classes	Mining (3), Metals (3), Commodity ETF (2), Equity Index (1), Bond (1)
Analysis Period	April 14, 2020 — April 14, 2025
Live Since	April 14, 2025
Transaction Cost Assumption	10 bps one-way per unit of turnover

Performance

5-Year Simulated Ann. Return	17.37%
5-Year Simulated Ann. Volatility	14.54%
5-Year In-Sample Sharpe	1.19
Rolling OOS Sharpe (net costs)	2.54
Live Return Apr 2025 — Apr 2026	+52.72%
Live Alpha vs SPY	+24.00 pp
Jensen's Alpha (5Y)	+7.06% (p=0.165)



Risk

Portfolio Beta vs SPY	0.50
Portfolio Beta vs VIX	-0.0025 (slight fear hedge)
Maximum Drawdown (5Y)	-18.67% (Apr 2022 — Jul 2023)
Calmar Ratio	0.93
Historical VaR 95%	-1.48% daily
Historical CVaR 95%	-2.06% daily
Historical VaR 99%	-2.25% daily
Historical CVaR 99%	-3.23% daily
Kupiec Test (95%)	PASS
Christoffersen Test (95%)	PASS
Current GARCH Vol State	NORMAL (~22% annualised)

Forecasting

Factor Model R-squared	93.9%
Top Factor	Silver (8.0% annual contribution)
HMM Current Regime	Bull (86.3% probability)
Copula MC Median 1Y Return	+20.86%
Copula MC 5th Percentile	-5.53%
Copula MC 95th Percentile	+51.78%
Probability of Positive Return	89.6%
Probability of Loss >10%	2.27%

Stress Test Results

Scenario	CasCom	SPY	Assessment
COVID Crash (Feb-Apr 2020)	~-25%	~-34%	Outperformed but known weakness (deflation)
2022 Rate Shock (Jan-Oct 2022)	+15%	-20%	Portfolio's design scenario; thesis fully validated
Tariff Shock (Apr 2025)	~-9%	~-14%	Origin scenario; recovered faster than SPY

Sensitivity Analysis Summary

Parameter	Finding	Conclusion
SHY Cap (3%-40%)	Sharpe flat at 1.19 across all caps	Cap is mandate choice, not performance driver
Weight Perturbation (±10%)	Max delta Sharpe: ±0.015	Robust to execution drift
Decay Lambda (0.985-0.999)	Rankings virtually frozen	Asset selection is lambda-independent
Correlation Cutoff (65%-85%)	10 assets survive above 80%	Universe is stable; TECK borderline at 75%

14. Appendix — Technical Details

14.1 Software & Libraries

Library	Purpose
yfinance	Price data download (Yahoo Finance API)
pandas / numpy	Data manipulation, returns calculation, matrix operations
scipy.optimize	Portfolio optimization (SLSQP method)
scipy.stats	Statistical tests (Kupiec, Christoffersen, t-distribution fitting)
arch	GARCH(1,1) fitting and walk-forward backtesting
hmmlearn	Hidden Markov Model regime switching
statsmodels	OLS regression (Factor Model, Jensen's Alpha)
matplotlib / seaborn	All charts and visualizations

14.2 Key Formulas

Capped Risk Parity Objective

Minimize the sum of squared deviations of each asset's risk contribution from the equal risk contribution target:

$\min \Sigma_i (RC_i - \sigma_p/n)^2$ subject to: $\Sigma w_i = 1$, $0.01 \le w_i \le 0.40$, $w(SHY) \le 0.06$

Exponential Decay Sharpe

$\lambda = 0.9945$ (6-month half-life)

$w_i = (1-\lambda)\cdot\lambda^i / \Sigma w_i$ (most recent day = highest weight)

Decay Sharpe = $(w^T r \cdot 252) / (\sqrt{(w^T \Sigma w)} \cdot \sqrt{252})$

GARCH(1,1)

$\sigma^2_t = \omega + \alpha\cdot\varepsilon^2_{t-1} + \beta\cdot\sigma^2_{t-1}$

Persistence = $\alpha + \beta$ (near 1.0 = permanent shocks)

Jensen's Alpha (OLS)

$(R_p - Rf) = \alpha + \beta\cdot(R_m - Rf) + \varepsilon$

α = annualised daily intercept × 252

Kupiec POF Test

$LR = -2[n \cdot \log(p) + (T-n) \cdot \log(1-p) - n \cdot \log(n/T) - (T-n) \cdot \log(1-(n/T))]$

$LR \sim \chi^2(1)$ under null of correct model

14.3 Data Sources

- All price data: Yahoo Finance via yfinance Python library (adjusted close prices)
- Risk-free rate: 13-week US Treasury Bill rate (^IRX) — average over analysis period
- VIX data: CBOE Volatility Index (^VIX) — daily changes used as fear proxy
- Analysis start: April 14, 2020 | Analysis end: April 14, 2025
- Live period: April 14, 2025 — April 14, 2026
- GARCH extended window: January 2018 — present (2 additional years for parameter stability)

14.4 Model Selection Rationale — Full Decision Log

Decision	Choice	Rationale
Weighting model	CRP	Best mandate-coherent OOS Sharpe; RP discipline with commodity tilt enforced by cap
Decay lambda	0.9945	6-month half-life — recent enough to penalise DBA/TLT weakness; long enough to avoid overreaction
SHY cap	6%	Provides a nominal liquidity/safety buffer without allowing bond allocation to dominate
Correlation threshold	75%	Standard academic threshold; borderline TECK and SLV kept with explicit justification
Transaction cost	10 bps	Conservative estimate covering ETF bid-ask spread; appropriate for quarterly rebalancing frequency
GARCH specification	(1,1)	Industry standard; additional lags add minimal predictive value at daily frequency
Copula type	t-Copula, df=4	Fat tails + tail dependence; captures correlation spike during stress better than Gaussian copula
HMM states	2	Two-state (Bull/Bear) sufficient for a commodity portfolio; additional states increase overfitting risk

AI Assistance Declaration

This research was conducted independently by Agshin Musayev. Artificial Intelligence tools (primarily Claude by Anthropic) were used in the following specific capacities:

What AI did:

- Generated all Python code for the project's Jupyter Notebook file.
- Produced the formatted layout and prose structure of this PDF document

What AI did not do:

- Define the investment thesis (tariff-driven inflation hedge via real assets)
- Select the asset universe or make inclusion/exclusion decisions
- Choose the specific models or determine the decision-process — these were original decisions made by the author
- Interpret results or draw investment conclusions
- Validate outputs against the live paper trading portfolio

Author's contribution: The portfolio thesis, asset selection rationale, all investment decisions, model selection logic, result interpretation, and live portfolio management are entirely the author's own work. Python and document generation tools were used as productivity instruments — equivalent to using Excel for financial modelling or Word for report writing — not as a substitute for investment judgment.

AI tools used: Claude (Anthropic) for code generation and document formatting. Data sourced from Yahoo Finance via yfinance.